

Make Your Web Great Again

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A. Konfigurasi Dasar

1. Buat user dengan Akses Sudo, Login menggunakan kata sandi dan menggunakan pubkey.

Name	Password
sevima-adm{1..1300}	w3bsite#1..1300 (buat password 1-1300 sesuai jumlah user)

Konfigurasi

a. Konfigurasi awal project dan buat file make_users.sh

```
ubuntu@ubuntu-2204:~$ mkdir Projects
ubuntu@ubuntu-2204:~$ cd Projects/
ubuntu@ubuntu-2204:~/Projects$ touch make_users.sh
```

b. Tambahkan script bash untuk otomatis pembuatan user sebagai berikut di file make_users.sh yang sudah dibuat:

```
ubuntu@ubuntu-2204:~/Projects$ cat make_users.sh
#!/bin/bash

for i in $(seq 1 1300); do
    sudo useradd -m -s /bin/bash "sevima-adm$i"
    echo "sevima-adm$i:w3bsite#$i" | sudo chpasswd
    sudo usermod -aG sudo "sevima-adm$i"
done
```

c. Buat bash script menjadi executable dan jalankan

```
ubuntu@ubuntu-2204:~/Projects$ chmod +x make_users.sh
ubuntu@ubuntu-2204:~/Projects$ ./make_users.sh
```

Validasi

Cek di /home apakah user-user sudah dibuat:

```
ubuntu@ubuntu-2204:/home$ ls
sevima-adm1  sevima-adm105  sevima-adm1212  sevima-adm15  sevima-adm257  sevima-adm364  sevima-adm471  sevima-adm579  sevima-adm686  sevima-adm793  sevima-adm90
sevima-adm10  sevima-adm106  sevima-adm1213  sevima-adm150  sevima-adm258  sevima-adm365  sevima-adm472  sevima-adm58  sevima-adm687  sevima-adm794  sevima-adm900
sevima-adm100  sevima-adm107  sevima-adm1214  sevima-adm151  sevima-adm259  sevima-adm366  sevima-adm473  sevima-adm580  sevima-adm688  sevima-adm795  sevima-adm901
sevima-adm1000  sevima-adm108  sevima-adm1215  sevima-adm152  sevima-adm26  sevima-adm367  sevima-adm474  sevima-adm581  sevima-adm689  sevima-adm796  sevima-adm902
sevima-adm1001  sevima-adm109  sevima-adm1216  sevima-adm153  sevima-adm260  sevima-adm368  sevima-adm475  sevima-adm582  sevima-adm69  sevima-adm797  sevima-adm903
sevima-adm1002  sevima-adm111  sevima-adm1217  sevima-adm154  sevima-adm261  sevima-adm369  sevima-adm476  sevima-adm583  sevima-adm690  sevima-adm798  sevima-adm904
sevima-adm1003  sevima-adm110  sevima-adm1218  sevima-adm155  sevima-adm262  sevima-adm37  sevima-adm477  sevima-adm584  sevima-adm691  sevima-adm799  sevima-adm905
sevima-adm1004  sevima-adm111  sevima-adm1219  sevima-adm156  sevima-adm263  sevima-adm370  sevima-adm478  sevima-adm585  sevima-adm692  sevima-adm8  sevima-adm906
sevima-adm1005  sevima-adm112  sevima-adm122  sevima-adm157  sevima-adm264  sevima-adm371  sevima-adm479  sevima-adm586  sevima-adm693  sevima-adm80  sevima-adm907
sevima-adm1006  sevima-adm113  sevima-adm1220  sevima-adm158  sevima-adm265  sevima-adm372  sevima-adm48  sevima-adm587  sevima-adm694  sevima-adm800  sevima-adm908
sevima-adm1007  sevima-adm114  sevima-adm1221  sevima-adm159  sevima-adm266  sevima-adm373  sevima-adm480  sevima-adm588  sevima-adm695  sevima-adm801  sevima-adm909
sevima-adm1008  sevima-adm115  sevima-adm1222  sevima-adm16  sevima-adm267  sevima-adm374  sevima-adm481  sevima-adm589  sevima-adm696  sevima-adm802  sevima-adm91
sevima-adm1009  sevima-adm116  sevima-adm1223  sevima-adm160  sevima-adm268  sevima-adm375  sevima-adm482  sevima-adm59  sevima-adm697  sevima-adm803  sevima-adm910
sevima-adm101  sevima-adm117  sevima-adm1224  sevima-adm161  sevima-adm269  sevima-adm376  sevima-adm483  sevima-adm590  sevima-adm698  sevima-adm804  sevima-adm911
sevima-adm1010  sevima-adm118  sevima-adm1225  sevima-adm162  sevima-adm27  sevima-adm377  sevima-adm484  sevima-adm591  sevima-adm699  sevima-adm805  sevima-adm912
sevima-adm1011  sevima-adm119  sevima-adm1226  sevima-adm163  sevima-adm270  sevima-adm378  sevima-adm485  sevima-adm592  sevima-adm7  sevima-adm806  sevima-adm913
sevima-adm1012  sevima-adm12  sevima-adm1227  sevima-adm164  sevima-adm271  sevima-adm379  sevima-adm486  sevima-adm593  sevima-adm70  sevima-adm807  sevima-adm914
sevima-adm1013  sevima-adm120  sevima-adm1228  sevima-adm165  sevima-adm272  sevima-adm38  sevima-adm487  sevima-adm594  sevima-adm700  sevima-adm808  sevima-adm915
sevima-adm1014  sevima-adm121  sevima-adm1229  sevima-adm166  sevima-adm273  sevima-adm380  sevima-adm488  sevima-adm595  sevima-adm701  sevima-adm809  sevima-adm916
sevima-adm1015  sevima-adm122  sevima-adm123  sevima-adm167  sevima-adm274  sevima-adm381  sevima-adm489  sevima-adm596  sevima-adm702  sevima-adm81  sevima-adm917
```

Cek apakah user 1 sampai 1300 sudah berhasil dibuat dengan passwordnya:

```
ubuntu@ubuntu-2204:~/Projects$ id sevima-adm1
uid=1001(sevima-adm1) gid=1001(sevima-adm1) groups=1001(sevima-adm1),27(sudo)
ubuntu@ubuntu-2204:~/Projects$ id sevima-adm1300
uid=2300(sevima-adm1300) gid=2300(sevima-adm1300) groups=2300(sevima-adm1300),27(sudo)
ubuntu@ubuntu-2204:~/Projects$ id sevima-adm777
uid=1777(sevima-adm777) gid=1777(sevima-adm777) groups=1777(sevima-adm777),27(sudo)
ubuntu@ubuntu-2204:~/Projects$ getent passwd sevima-adm1
sevima-adm1:x:1001:1001::/home/sevima-adm1:/bin/bash
ubuntu@ubuntu-2204:~/Projects$ getent passwd sevima-adm1300
sevima-adm1300:x:2300:2300::/home/sevima-adm1300:/bin/bash
ubuntu@ubuntu-2204:~/Projects$ getent passwd sevima-adm777
sevima-adm777:x:1777:1777::/home/sevima-adm777:/bin/bash
```

- d. Buat backup file sshd_config terlebih dahulu

```
ubuntu@ubuntu-2204:/etc/ssh$ sudo cp sshd_config sshd_config.bak
ubuntu@ubuntu-2204:/etc/ssh$ ls
moduli          ssh_config.d  sshd_config.bak  ssh_host_ecdsa_key  ssh_host_ed25519_key
ssh_config      sshd_config  sshd_config.d    ssh_host_ecdsa_key.pub  ssh_host_ed25519_key.pub
```

- e. Enable pubkey di sshd_config (jangan disable PasswordAuthentication sesuai permintaan soal)

```
PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
#PermitEmptyPasswords no

# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing. If this is enabled, PAM authentication will
# be allowed through the KbdInteractiveAuthentication and
# PasswordAuthentication. Depending on your PAM configuration,
# PAM authentication via KbdInteractiveAuthentication may bypass
# the setting of "PermitRootLogin without-password".
# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes
```

- f. Test syntax

```
ubuntu@ubuntu-2204:/etc/ssh$ sudo sshd -t
ubuntu@ubuntu-2204:/etc/ssh$
```

- g. Setelah dipastikan bahwa tidak ada masalah, restart service sshd

```
ubuntu@ubuntu-2204:/etc/ssh$ sudo systemctl restart sshd
ubuntu@ubuntu-2204:/etc/ssh$ sudo systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-09-21 10:50:33 EDT; 2s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 37466 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 37467 (sshd)
    Tasks: 1 (limit: 4923)
   Memory: 1.7M
      CPU: 19ms
   CGroup: /system.slice/ssh.service
           └─37467 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

Validasi

Login menggunakan password:

```
(kali㉿kali)-[~]
└─$ ssh sevima-adm777@192.168.34.135
The authenticity of host '192.168.34.135 (192.168.34.135)' can't be established.
ED25519 key fingerprint is: SHA256:sSTOXKPPMFids0dDVld8mUsZFhWxyK8HVilZq1EcXGM
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.34.135' (ED25519) to the list of known hosts.
+-----+
                        LINUXVMIMAGES.COM
+-----+
User Name: ubuntu
Password: ubuntu (sudo su -)
sevima-adm777@192.168.34.135's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

0 updates can be applied immediately.

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
+-----+
                        LINUXVMIMAGES.COM
+-----+
User Name: ubuntu
Password: ubuntu (sudo su -)
Last login: Mon Sep 21 10:31:37 2026 from 192.168.34.134
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

sevima-adm777@ubuntu-2204:~$ uname -a
Linux ubuntu-2204 5.15.0-25-generic #25-Ubuntu SMP Wed Mar 30 15:54:22 UTC 2022 x86_64 x86_64
x86_64 GNU/Linux
```

Login menggunakan PubKey:

- Generate key di machine yang ingin login:

```
(kali㉿kali)-[~]
└─$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/kali/.ssh/id_ed25519):
Enter passphrase for "/home/kali/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/kali/.ssh/id_ed25519
Your public key has been saved in /home/kali/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:sAtCAkQwGrkbs6cG0BeC2h/Y2hcXZU10AC+w1ecNcY4 kali@kali
The key's randomart image is:
+--[ED25519 256]--+
|O+      . o**o..|
|+o      +o.o.o+ |
|+..     .o. .oEo.|
|*o.     o. . . .|
|oB+.    . S.    |
|B =.o....|
|.= + ..o|
|... o .|
|. . .|
+---[SHA256]-----+
```

- Copy public key ke server ubuntu:

```
(kali㉿kali)-[~]
└─$ ssh-copy-id sevima-adm777@192.168.34.135
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/kali/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that
are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is
to install the new keys
+-----+
|               LINUXVMIMAGES.COM               |
+-----+
User Name: ubuntu
Password: ubuntu (sudo su -)
sevima-adm777@192.168.34.135's password:
Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'sevima-adm777@192.168.34.135'"
and check to make sure that only the key(s) you wanted were added.
```

- Cek apakah public key sudah dibuat di server ubuntu:

```
sevima-adm777@ubuntu-2204:~/.ssh$ ls -la
total 12
drwx----- 2 sevima-adm777 sevima-adm777 4096 Sep 21 11:03 .
drwxr-x--- 7 sevima-adm777 sevima-adm777 4096 Sep 21 11:03 ..
-rw----- 1 sevima-adm777 sevima-adm777 91 Sep 21 11:03 authorized_keys
sevima-adm777@ubuntu-2204:~/.ssh$ cat authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIJ5KF4l7NYq3Y1WcRdaq2fbUv0ZAwk3F5QLMDNMfr0Ma kali@kali
sevima-adm777@ubuntu-2204:~/.ssh$ whoami
sevima-adm777
```

- Coba login dari kali:

```
(kali㉿kali)-[~]
└─$ ssh sevima-adm777@192.168.34.135
+-----+
|               LINUXVMIMAGES.COM               |
+-----+
User Name: ubuntu
Password: ubuntu (sudo su -)
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

0 updates can be applied immediately.

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

+-----+
|               LINUXVMIMAGES.COM               |
+-----+
User Name: ubuntu
Password: ubuntu (sudo su -)
Last login: Mon Sep 21 10:52:33 2026 from 192.168.34.134
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

sevima-adm777@ubuntu-2204:~$ whoami
sevima-adm777
sevima-adm777@ubuntu-2204:~$ uname -a
Linux ubuntu-2204 5.15.0-25-generic #25-Ubuntu SMP Wed Mar 30 15:54:22 UTC 2022 x86_64 x86_64
x86_64 GNU/Linux
```

Bisa dilihat bahwa kita tidak perlu lagi login dengan password, cukup dengan pubkey.

2. Ubah port ssh menjadi 2025

Konfigurasi

- Ubah konfigurasi port ssh

```
Port 2025
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
```

b. Restart service sshd

```
ubuntu@ubuntu-2204:/etc/ssh$ sudo systemctl restart sshd
ubuntu@ubuntu-2204:/etc/ssh$ sudo systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-09-21 10:01:27 EDT; 4s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Process: 2820 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
 Main PID: 2821 (sshd)
    Tasks: 1 (limit: 4923)
   Memory: 1.7M
      CPU: 33ms
   CGroup: /system.slice/ssh.service
           └─2821 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

Validasi

Before:

```
(kali@kali)-[~]
$ nmap -Pn -sV -O -T4 192.168.34.135
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-21 20:47 +0700
Nmap scan report for 192.168.34.135
Host is up (0.00060s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3 (Ubuntu Linux; protocol 2.0)
MAC Address: 00:0C:29:7A:FB:F0 (VMware)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

After:

```
(kali@kali)-[~]
$ nmap -Pn -sV -O -p- -T4 192.168.34.135
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-21 21:03 +0700
Nmap scan report for 192.168.34.135
Host is up (0.00084s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
2025/tcp  open  ssh      OpenSSH 8.9p1 Ubuntu 3 (Ubuntu Linux; protocol 2.0)
MAC Address: 00:0C:29:7A:FB:F0 (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

3. Aktifkan seluruh log aktivitas

Konfigurasi

- Buat direktori /var/log/journal dengan `sudo mkdir -p /var/log/journal`
- Ubah Storage dari auto menjadi persistent

```
[Journal]
Storage=persistent
#Compress=yes
#Seal=yes
#SplitMode=uid
#SyncIntervalSec=5m
#RateLimitIntervalSec=30s
#RateLimitBurst=10000
#SystemMaxUse=
#SystemKeepFree=
#SystemMaxFileSize=
```

- c. Restart systemd-journald dengan `sudo systemctl restart systemd-journald`

```
sevima-adm777@ubuntu-2204:/var/log$ sudo systemctl restart systemd-journald
```

Validasi

Cek apakah service journald berjalan dan storagenya:

```
sevima-adm777@ubuntu-2204:/var/log$ sudo journalctl --disk-usage
Archived and active journals take up 96.0M in the file system.
sevima-adm777@ubuntu-2204:/var/log$ sudo systemctl status systemd-journald
● systemd-journald.service - Journal Service
   Loaded: loaded (/lib/systemd/system/systemd-journald.service; static)
   Active: active (running) since Mon 2026-09-21 11:23:14 EDT; 1min 25s ago
 TriggeredBy: ● systemd-journald-audit.socket
               ● systemd-journald.socket
               ● systemd-journald-dev-log.socket
   Docs: man:systemd-journald.service(8)
         man:journald.conf(5)
 Main PID: 39021 (systemd-journal)
  Status: "Processing requests..."
   Tasks: 1 (limit: 4923)
  Memory: 1.4M
    CPU: 91ms
   CGroup: /system.slice/systemd-journald.service
           └─39021 /lib/systemd/systemd-journald
```

4. Sesuaikan penggunaan sumber daya sesuai kebutuhan dengan ulimit

Konfigurasi

- a. tambahkan rules ulimit pada /etc/security/limits.conf

```
# End of file

* soft nfile 65535
* hard nfile 65535

* soft nproc 8192
* hard nproc 8192

root soft nfile 65535
root hard nfile 65535
```

Validasi

logout dan login Kembali, kemudian cek ulimit yang sudah diupdate

```
root@ubuntu-2204:/etc/security# su -
root@ubuntu-2204:~# ^C
root@ubuntu-2204:~# ulimit -n
65535
```

B. Certificate Authority

1. Set up root certificate authority menggunakan OpenSSL pada direktori /root/ca.

Konfigurasi

a. Buat directory:

```
sevima-adm777@ubuntu-2204:~$ sudo mkdir -p /root/ca
sevima-adm777@ubuntu-2204:~$ sudo chmod 700 /root/ca
sevima-adm777@ubuntu-2204:~$ ls -la /root
ls: cannot open directory '/root': Permission denied
sevima-adm777@ubuntu-2204:~$ sudo ls -la /root
total 36
drwx----- 6 root root 4096 Sep 21 22:48 .
drwxr-xr-x 20 root root 4096 Apr 22 2022 ..
-rw-r--r-- 1 root root 3106 Oct 15 2021 .bashrc
drwx----- 2 root root 4096 Sep 21 22:48 ca
drwx----- 2 root root 4096 Apr 19 2022 .cache
-rw----- 1 root root 20 Sep 21 11:24 .lessht
drwxr-xr-x 3 root root 4096 Apr 22 2022 .local
-rw-r--r-- 1 root root 161 Jul 9 2019 .profile
drwx----- 5 root root 4096 Apr 22 2022 snap
```

b. Login sebagai root dengan `sudo su` dan pergi ke `/root/ca`

c. Generate CA private key:

```
root@ubuntu-2204:~/ca# openssl genrsa -out cacert.key 4096
root@ubuntu-2204:~/ca# ls
cacert.key
root@ubuntu-2204:~/ca# cat cacert.key
-----BEGIN PRIVATE KEY-----
MIIJQgIBADANBgkqhkiG9w0BAQEFAASCCSwggkoAgEAAoICAQDMbN8l/5vsWUbm
jNTlCcKaVmftVo8mpy3d3gvdCwLhJ0RUgnAsD/lADgEifkBWk7BLR6KUV0A6jz40
S207ul5luD5XERRo0HLY0xyhMzR/UNkttcCz52AP5ZIs+LfLkToDPJXet+qAwX2B
Gb8itUMWM1BkUG7D+I6Ums3nRUMyTbS/g+aH1MbAwJjL7goyN51dMhEz7lCojCNX
Hu//pwbmoXCCQjyVC6h63lFsnSBpHh+tyszG7MgYr0aXv3M5K4pLRcAFW/mqQz9L
```

d. Generate root cert:

```
root@ubuntu-2204:~/ca# openssl req -x509 -new -key cacert.key -sha256 -days 3650 -out cacert.pem -subj "/C=ID/O=PT. Sentra Vidya Utama/CN=SEVIMA CA"
root@ubuntu-2204:~/ca# openssl -x509 -in cacert.pem -noout -subject -issuer -dates
Invalid command '-x509': type "help" for a list.
root@ubuntu-2204:~/ca# openssl x509 -in cacert.pem -noout -subject -issuer -dates
subject=C = ID, O = PT. Sentra Vidya Utama, CN = SEVIMA CA
issuer=C = ID, O = PT. Sentra Vidya Utama, CN = SEVIMA CA
notBefore=Sep 22 02:57:17 2026 GMT
notAfter=Sep 19 02:57:17 2036 GMT
```

e. Setup permission

```
root@ubuntu-2204:~/ca# chmod 600 cacert.key
root@ubuntu-2204:~/ca# chmod 644 cacert.pem
root@ubuntu-2204:~/ca# ls -la
total 16
drwx----- 2 root root 4096 Sep 21 22:57 .
drwx----- 6 root root 4096 Sep 21 22:48 ..
-rw----- 1 root root 3272 Sep 21 22:51 cacert.key
-rw-r--r-- 1 root root 1931 Sep 21 22:57 cacert.pem
```

2. Buat Root Certificate cacert.pem and cacert.key dengan menggunakan informasi di bawah ini:

a) Country code: ID

b) Organization: PT. Sentra Vidya Utama

c) Common Name: SEVIMA CA

d) Create additional certificate

Issued Certificate	Note
www.sevima.site	Web
utara.sevima.site	Web
timur.sevima.site	Web
barat.sevima.site	Web

Konfigurasi

a. setup certificate utara.sevima.site

```
root@ubuntu-2204:~/ca# openssl genrsa -out utara.sevima.site.key 2048
root@ubuntu-2204:~/ca# openssl req -new \
-key utara.sevima.site.key \
-out utara.sevima.site.csr \
-subj "/C=ID/O=PT. Sentra Vidya Utama/CN=utara.sevima.site"
root@ubuntu-2204:~/ca# ls
cacert.key  cacert.pem  utara.sevima.site.csr  utara.sevima.site.key
```

Buat SAN config dan sign:

```
root@ubuntu-2204:~/ca# cat utara.ext
authorityKeyIdentifier=keyid,issuer
basicConstraints=CA:FALSE
keyUsage=digitalSignature,keyEncipherment
extendedKeyUsage=serverAuth
subjectAltName=@alt_names

[alt_names]
DNS.1=utara.sevima.site
root@ubuntu-2204:~/ca# openssl x509 -req \
-in utara.sevima.site.csr \
-CA cacert.pem \
-CAkey cacert.key \
-CAcreateserial \
-out utara.sevima.site.crt \
-days 825 \
-sha256 \
-extfile utara.ext
Certificate request self-signature ok
subject=C = ID, O = PT. Sentra Vidya Utama, CN = utara.sevima.site
```

b. setup certificate timur.sevima.site

```
root@ubuntu-2204:~/ca# openssl genrsa -out timur.sevima.site.key 2048
root@ubuntu-2204:~/ca# openssl req -new \
-key timur.sevima.site.key \
-out timur.sevima.site.csr \
-subj "/C=ID/O=PT. Sentra Vidya Utama/CN=timur.sevima.site"
root@ubuntu-2204:~/ca# ls
cacert.key  cacert.pem  timur.sevima.site.csr  timur.sevima.site.key  utara.ext
```

Buat SAN config dan sign:

```
root@ubuntu-2204:~/ca# touch timur.ext
root@ubuntu-2204:~/ca# nano timur.ext
root@ubuntu-2204:~/ca# openssl x509 -req \
-in timur.sevima.site.csr \
-CA cacert.pem \
-CAkey cacert.key \
-CAserial cacert.srl \
-out timur.sevima.site.crt \
-days 825 \
-sha256 \
-extfile timur.ext
Certificate request self-signature ok
subject=C = ID, O = PT. Sentra Vidya Utama, CN = timur.sevima.site
```

c. setup certificate barat.sevima.site

```
root@ubuntu-2204:~/ca# openssl genrsa -out barat.sevima.site.key 2048
root@ubuntu-2204:~/ca# openssl req -new \
-key barat.sevima.site.key \
-out barat.sevima.site.csr \
-subj "/C=ID/O=PT. Sentra Vidya Utama/CN=barat.sevima.site"
root@ubuntu-2204:~/ca# ls | grep barat
barat.sevima.site.csr
barat.sevima.site.key
```

Buat SAN config dan sign:

```
root@ubuntu-2204:~/ca# touch barat.ext
root@ubuntu-2204:~/ca# nano barat.ext
root@ubuntu-2204:~/ca# openssl x509 -req \
-in barat.sevima.site.csr \
-CA cacert.pem \
-CAkey cacert.key \
-CAserial cacert.srl \
-out barat.sevima.site.crt \
-days 825 \
-sha256 \
-extfile barat.ext
Certificate request self-signature ok
subject=C = ID, O = PT. Sentra Vidya Utama, CN = barat.sevima.site
```

d. setup certificate www.sevima.site

```
root@ubuntu-2204:~/ca# openssl genrsa -out www.sevima.site.key 2048
root@ubuntu-2204:~/ca# openssl req -new \
-key www.sevima.site.key \
-out www.sevima.site.csr \
-subj "/C=ID/O=PT. Sentra Vidya Utama/CN=www.sevima.site"
```

Buat SAN config dan sign:

```

root@ubuntu-2204:~/ca# touch www.ext
root@ubuntu-2204:~/ca# nano www.ext
root@ubuntu-2204:~/ca# openssl x509 -req \
-in www.sevima.site.csr \
-CA cacert.pem \
-CAkey cacert.key \
-CAserial cacert.srl \
-out www.sevima.site.crt \
-days 825 \
-sha256 \
-extfile www.ext
Certificate request self-signature ok
subject=C = ID, O = PT. Sentra Vidya Utama, CN = www.sevima.site

```

Validasi

Verifikasi keempathnya:

```

root@ubuntu-2204:~# openssl verify -CAfile /root/ca/cacert.pem /root/ca/utara.sevima.site.crt
/root/ca/utara.sevima.site.crt: OK
root@ubuntu-2204:~# openssl verify -CAfile /root/ca/cacert.pem /root/ca/timur.sevima.site.crt
/root/ca/timur.sevima.site.crt: OK
root@ubuntu-2204:~# openssl verify -CAfile /root/ca/cacert.pem /root/ca/barat.sevima.site.crt
/root/ca/barat.sevima.site.crt: OK
root@ubuntu-2204:~# openssl verify -CAfile /root/ca/cacert.pem /root/ca/www.sevima.site.crt
/root/ca/www.sevima.site.crt: OK

```

C. Web Server

a. Install web server menggunakan apache2 and nginx di host yang sama.

1. Buat virtual host HTTP-only untuk melayani utara.sevima.site menggunakan apache2 pada port 80 + ID peserta Contoh: jika ID peserta adalah 23, maka gunakan port 8023. (dari nusalearn.sevima.com, ID saya adalah 29)

a) Halaman website harus menampilkan "Hello World from Utara Site"

b) Tambahkan HTTP header berikut:

- X-Owner-By: diisi dengan nama peserta

- X-Served-By: diisi dengan jenis web server (apache2)

Konfigurasi

a. Install apache2

```

sevima-adm777@ubuntu-2204:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap
0 upgraded, 8 newly installed, 0 to remove and 724 not upgraded.
Need to get 1,937 kB of archives.
After this operation, 7,767 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [108 kB]
Get:2 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.3 [93.4 kB]
Get:3 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.3 [11.4 kB]
Get:4 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.3 [9,164 B]

```

b. Matikan / disable default site:

```

sevima-adm777@ubuntu-2204:/etc/apache2$ sudo a2dissite 000-default.conf
Site 000-default disabled.
To activate the new configuration, you need to run:
  systemctl reload apache2
sevima-adm777@ubuntu-2204:/etc/apache2$ sudo systemctl reload apache2

```

c. Konfigurasi port sesuai ID peserta (saya menggunakan ID 29) di `/etc/apache2/ports.conf`

```

Listen 8029

<IfModule ssl_module>
    Listen 443
</IfModule>

<IfModule mod_gnutls.c>
    Listen 443
</IfModule>

```

- d. Buat halaman website di `/var/www/utara/index.html`

```
root@ubuntu-2204:/etc/apache2# cd /var/www/
root@ubuntu-2204:/var/www# ls
html
root@ubuntu-2204:/var/www# mkdir utara
root@ubuntu-2204:/var/www# cd utara/
root@ubuntu-2204:/var/www/utara# touch index.html
root@ubuntu-2204:/var/www/utara# nano index.html
root@ubuntu-2204:/var/www/utara# cat index.html
<!DOCTYPE html>
<html>
<head>
  <title>Utara Sevima</title>
</head>
<body>
  <h1>Hello World from Utara Site</h1>
</body>
</html>
```

- e. Buat virtual host di `/etc/apache2/sites-available/utara.sevima.site.conf`

```
<VirtualHost *:8029>
    ServerName utara.sevima.site

    DocumentRoot /var/www/utara

    <Directory /var/www/utara>
        AllowOverride None
        Require all granted
    </Directory>

    Header always set X-Owner-By "YEHEZKIEL WIRADHIKA"
    Header always set X-Served-By "apache2"

    ErrorLog ${APACHE_LOG_DIR}/utara_error.log
    CustomLog ${APACHE_LOG_DIR}/utara_access.log combined
</VirtualHost>
```

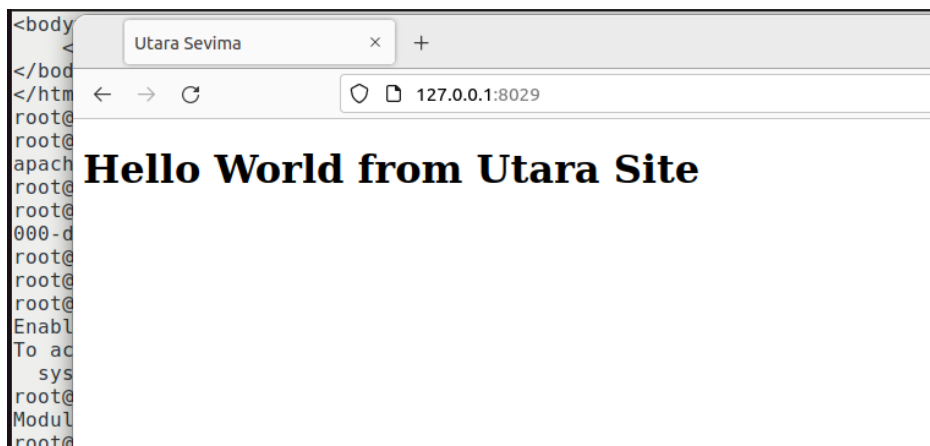
- f. Aktifkan module header, enable site dan test

```
root@ubuntu-2204:/etc/apache2/sites-available# sudo a2enmod headers
Enabling module headers.
To activate the new configuration, you need to run:
  systemctl restart apache2
root@ubuntu-2204:/etc/apache2/sites-available# a2enmod headers
Module headers already enabled
root@ubuntu-2204:/etc/apache2/sites-available# a2ensite utara.sevima.site.conf
Enabling site utara.sevima.site.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@ubuntu-2204:/etc/apache2/sites-available# apache2ctl configtest
Syntax OK
```

- g. Restart apache dengan `sudo systemctl reload apache2`

Validasi

Coba akses dengan 127.0.0.1 (localhost) di browser ubuntu:



2. Buat virtual host HTTP-only untuk melayani timur.sevima.site menggunakan nginx pada port 81 + ID peserta Contoh: jika ID peserta adalah 23, maka gunakan port 8123.

a) Halaman website harus menampilkan "Hello World from Timur Site".

b) Tambahkan HTTP header berikut:

```

root@ubuntu-2204:/var/www/timur# cd /etc/nginx/sites-available/
root@ubuntu-2204:/etc/nginx/sites-available# touch timur.sevima.site
root@ubuntu-2204:/etc/nginx/sites-available# nano timur.sevima.site
root@ubuntu-2204:/etc/nginx/sites-available# ln -s /etc/nginx/sites-available/timur.sevima.site /etc/nginx/sites-enabled/
root@ubuntu-2204:/etc/nginx/sites-available# nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
root@ubuntu-2204:/etc/nginx/sites-available# systemctl restart nginx
root@ubuntu-2204:/etc/nginx/sites-available# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-09-22 10:16:24 EDT; 5s ago
     Docs: man:nginx(8)
  Process: 50657 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Process: 50658 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
 Main PID: 50659 (nginx)
    Tasks: 3 (limit: 4923)
   Memory: 3.3M
      CPU: 157ms
  CGroup: /system.slice/nginx.service
          └─50659 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
            └─50660 "nginx: worker process"
              └─50661 "nginx: worker process"

Sep 22 10:16:23 ubuntu-2204 systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 22 10:16:24 ubuntu-2204 systemd[1]: Started A high performance web server and a reverse proxy server.

```

Validasi

Coba akses dengan 127.0.0.1 (localhost) di browser ubuntu:



3. Buat virtual host HTTPS untuk melayani barat.sevima.site menggunakan nginx dengan port https angka acak (contoh: 4435)

- a) Aktifkan HTTPS menggunakan Certificate Authority dari SEVIMA CA.
- b) Redirect semua permintaan HTTP ke HTTPS.
- c) Pastikan localhost dan laptop kamu dapat mengakses tanpa peringatan apa pun.

Konfigurasi

- a. Buat konten website di `/var/www/barat/index.html`

```
<!DOCTYPE html>
<html>
<head>
  <title>Barat Sevima</title>
</head>
<body>
  <h1>Hello World from Barat Site</h1>
</body>
</html>
```

- b. Copy certificate

```
root@ubuntu-2204:/var/www/barat# sudo mkdir -p /etc/nginx/ssl
root@ubuntu-2204:/var/www/barat# sudo cp /root/ca/barat.sevima.site.crt /etc/nginx/ssl/
root@ubuntu-2204:/var/www/barat# sudo cp /root/ca/barat.sevima.site.key /etc/nginx/ssl/
root@ubuntu-2204:/var/www/barat# sudo chmod 600 /etc/nginx/ssl/barat.sevima.site.key
root@ubuntu-2204:/var/www/barat# ls -la /etc/nginx/ssl/ | grep barat
-rw----- 1 root root 1675 Sep 22 10:26 barat.sevima.site.crt
-rw----- 1 root root 1704 Sep 22 10:26 barat.sevima.site.key
```

- c. Buat konfigurasi file `/etc/nginx/sites-available/barat.sevima.site`

```
server {
    listen 8080;
    listen [::]:8080;

    server_name barat.sevima.site;

    return 301 https://barat.sevima.site:8443$request_uri;
}

server {
    listen 8443 ssl;
    listen [::]:8443 ssl;

    server_name barat.sevima.site;

    root /var/www/barat;
    index index.html;

    ssl_certificate /etc/nginx/ssl/barat.sevima.site.crt;
    ssl_certificate_key /etc/nginx/ssl/barat.sevima.site.key;

    ssl_protocols TLSv1.2 TLSv1.3;

    add_header X-Owner-By "Yehezkiel Wiradhika" always;
    add_header X-Served-By "nginx" always;

    location / {
        try_files $uri $uri/ =404;
    }

    access_log /var/log/nginx/barat_access.log;
    error_log /var/log/nginx/barat_error.log;
}
```

- d. Enable, test dan restart nginx

```
root@ubuntu-2204:/etc/nginx/sites-available# sudo ln -s /etc/nginx/sites-available/barat.sevima.site /etc/nginx/sites-enabled/
root@ubuntu-2204:/etc/nginx/sites-available# nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
root@ubuntu-2204:/etc/nginx/sites-available# systemctl restart nginx
root@ubuntu-2204:/etc/nginx/sites-available# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-09-22 10:30:40 EDT; 2s ago
     Docs: man:nginx(8)
```

- e. Aktivasi hosts untuk local di `/etc/hosts`

```
GNU nano 6.2 /etc/hosts
127.0.0.1 localhost
127.0.1.1 ubuntu-2204.linuxvmimages.local ubuntu-2204

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

127.0.0.1 utara.sevima.site
127.0.0.1 timur.sevima.site
127.0.0.1 barat.sevima.site
127.0.0.1 www.sevima.site
```

- f. Konfigurasi agar bisa diakses dari luar (kali linux – di `/etc/hosts`)

```
(kali@kali)~[~/Projects/nikto]
$ cat /etc/hosts
127.0.0.1 localhost
127.0.1.1 kali
::1 localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

192.168.34.135 utara.sevima.site
192.168.34.135 timur.sevima.site
192.168.34.135 barat.sevima.site
192.168.34.135 selatan.sevima.site
```

Validasi

Cek apakah hosts sudah teraktivasi

```

root@ubuntu-2204:/etc/nginx/sites-available# getent hosts utara.sevima.site
getent hosts timur.sevima.site
getent hosts barat.sevima.site
getent hosts www.sevima.site
127.0.0.1      utara.sevima.site
127.0.0.1      timur.sevima.site
127.0.0.1      barat.sevima.site
127.0.0.1      www.sevima.site
root@ubuntu-2204:/etc/nginx/sites-available# curl --cacert /root/ca/cacert.pem https://barat.sevima.site:8443
<!DOCTYPE html>
<html>
<head>
  <title>Barat Sevima</title>
</head>
<body>
  <h1>Hello World from Barat Site</h1>
</body>
</html>

```

Cek menggunakan localhost (redirect Ketika mengakses <http://127.0.0.1:8080>)

```

root@ubuntu-2204:/etc/nginx/sites-available# curl http://127.0.0.1:8080
<html>
<head><title>301 Moved Permanently</title></head>
<body>
<center><h1>301 Moved Permanently</h1></center>
<hr><center>nginx/1.18.0 (Ubuntu)</center>
</body>
</html>

```

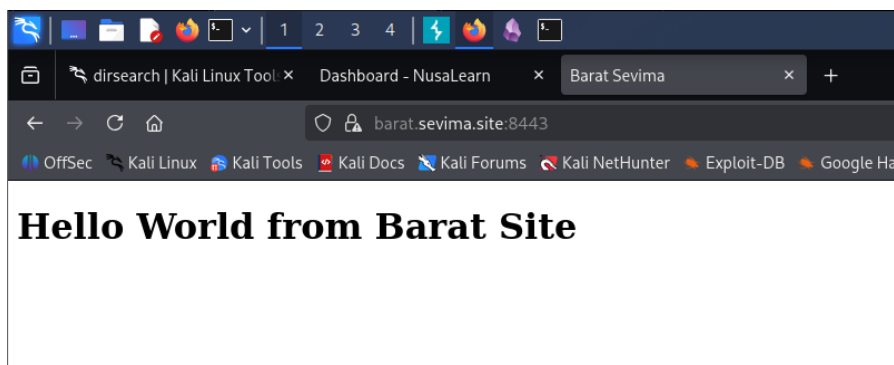


Cek menggunakan kali linux (machine lain di network yang sama)

```

(kali@kali)-[~]
$ curl http://barat.sevima.site:8080
<html>
<head><title>301 Moved Permanently</title></head>
<body>
<center><h1>301 Moved Permanently</h1></center>
<hr><center>nginx/1.18.0 (Ubuntu)</center>
</body>
</html>

```



D. Load Balancer HAProxy

- 1) Konfigurasi load balancer HTTP/HTTPS untuk www.sevima.site, yang dihosting oleh utara.sevima.site dan timur.sevima.site menggunakan apache2 dan nginx.
- 2) Gunakan sertifikat dari SEVIMA CA.

3) Gunakan algoritma round-robin.

4) Pastikan localhost dapat mengakses tanpa peringatan apa pun.

Konfigurasi

a. Install haproxy

```
root@ubuntu-2204:~# apt install haproxy -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  vim-haproxy haproxy-doc
The following NEW packages will be installed:
  haproxy
0 upgraded, 1 newly installed, 0 to remove and 723 not upgraded.
Need to get 1,655 kB of archives.
After this operation, 3,719 kB of additional disk space will be used
Get:1 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 haproxy amd64 2.8.0-2ubuntu0.2 [1,655 kB]
debconf: delaying package configuration, since apt-utils is not installed
```

b. Buat cert untuk haproxy

```
root@ubuntu-2204:~# sudo mkdir -p /etc/haproxy/certs
root@ubuntu-2204:~# sudo sh -c 'cat /root/ca/www.sevima.site.crt /root/ca/www.sevima.site.key > /etc/haproxy/certs/www.sevima.site.pem'
root@ubuntu-2204:~# sudo chmod 600 /etc/haproxy/certs/www.sevima.site.pem
```

c. Konfigurasi haproxy di `/etc/haproxy/haproxy.cfg`

```
global
    log /dev/log local0
    log /dev/log local1 notice

    maxconn 4096

    user haproxy
    group haproxy

    daemon

defaults
    log global

    mode http

    option httplog
    option dontlognull

    timeout connect 5s
    timeout client 30s
    timeout server 30s

frontend www http
    bind *:80

    http-request redirect scheme https code 301

frontend www https
    bind *:443 ssl crt /etc/haproxy/certs/www.sevima.site.pem

    mode http

    default_backend sevima_web

backend sevima_web
    balance roundrobin

    option httpchk GET /

    server utara 127.0.0.1:8029 check
    server timur 127.0.0.1:8129 check
```

d. Cek, restart dan enable

```
root@ubuntu-2204:/etc/haproxy# sudo haproxy -c -f /etc/haproxy/haproxy.cfg
Configuration file is valid
root@ubuntu-2204:/etc/haproxy# sudo systemctl restart haproxy
root@ubuntu-2204:/etc/haproxy# sudo systemctl enable haproxy
Synchronizing state of haproxy.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable haproxy
root@ubuntu-2204:/etc/haproxy# nano haproxy.cfg
root@ubuntu-2204:/etc/haproxy# sudo systemctl status haproxy
● haproxy.service - HAProxy Load Balancer
   Loaded: loaded (/lib/systemd/system/haproxy.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-09-22 23:47:18 EDT; 58s ago
     Docs: man:haproxy(1)
           file:/usr/share/doc/haproxy/configuration.txt.gz
  Main PID: 54950 (haproxy)
    Tasks: 3 (limit: 4923)
   Memory: 9.3M
      CPU: 199ms
   CGroup: /system.slice/haproxy.service
           └─54950 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid
           └─54952 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid
```

Validasi

Cek port dan services yang berjalan dengan command `ss -lntp`

```
root@ubuntu-2204:/etc/security# ss -lntp | grep -E ":2025|:80|:443|:8029|:8129|:8080|:8443"
LISTEN 0      128          0.0.0.0:*      0.0.0.0:2025      users: ("sshd",pid=55102,fd=3))
LISTEN 0      4096         0.0.0.0:80      0.0.0.0:*        users: ("haproxy",pid=56424,fd=6))
LISTEN 0      511         0.0.0.0:8080    0.0.0.0:*        users: ("nginx",pid=56179,fd=10),("nginx",pid=56178,fd=10),("nginx",pid=56177,fd=10))
LISTEN 0      4096         0.0.0.0:443     0.0.0.0:*        users: ("haproxy",pid=56424,fd=7))
LISTEN 0      511         0.0.0.0:8443    0.0.0.0:*        users: ("nginx",pid=56179,fd=12),("nginx",pid=56178,fd=12),("nginx",pid=56177,fd=12))
LISTEN 0      511         0.0.0.0:8129    0.0.0.0:*        users: ("nginx",pid=56179,fd=14),("nginx",pid=56178,fd=14),("nginx",pid=56177,fd=14))
LISTEN 0      128         [::]:2025      [::]:*          users: ("sshd",pid=55102,fd=4))
LISTEN 0      511         [::]:8080      [::]:*          users: ("nginx",pid=56179,fd=11),("nginx",pid=56178,fd=11),("nginx",pid=56177,fd=11))
LISTEN 0      511         [::]:8443      [::]:*          users: ("nginx",pid=56179,fd=13),("nginx",pid=56178,fd=13),("nginx",pid=56177,fd=13))
LISTEN 0      511         *:8029         *:*             users: ("apache2",pid=55227,fd=7),("apache2",pid=55226,fd=7),("apache2",pid=47423,fd=7))
LISTEN 0      511         [::]:8129      [::]:*          users: ("nginx",pid=56179,fd=15),("nginx",pid=56178,fd=15),("nginx",pid=56177,fd=15))
```

Cek certificate haproxy

```
root@ubuntu-2204:/etc/haproxy# openssl s_client \
-connect www.sevima.site:443 \
-servername www.sevima.site
CONNECTED(00000003)
depth=1 C = ID, O = PT. Sentra Vidya Utama, CN = SEVIMA CA
verify return:1
depth=0 C = ID, O = PT. Sentra Vidya Utama, CN = www.sevima.site
verify return:1
---
Certificate chain
 0 s:C = ID, O = PT. Sentra Vidya Utama, CN = www.sevima.site
  i:C = ID, O = PT. Sentra Vidya Utama, CN = SEVIMA CA
  a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
  v:NotBefore: Sep 22 14:42:45 2026 GMT; NotAfter: Dec 25 14:42:45 2028 GMT
```

Cek roundrobin

```
root@ubuntu-2204:~# cat test_roundrobin.sh
#!/bin/bash

for i in {1..10}; do
    curl -sk https://www.sevima.site
    echo
done
```

```
root@ubuntu-2204:~# ./test_roundrobin.sh | grep -oE "<h1>.*</h1>"
<h1>Hello World from Utara Site</h1>
<h1>Hello World from Timur Site</h1>
<h1>Hello World from Utara Site</h1>
<h1>Hello World from Timur Site</h1>
<h1>Hello World from Utara Site</h1>
<h1>Hello World from Timur Site</h1>
<h1>Hello World from Utara Site</h1>
<h1>Hello World from Timur Site</h1>
<h1>Hello World from Utara Site</h1>
<h1>Hello World from Timur Site</h1>
```